

devices for clicking photographs in the past three years. Smartphone cameras have almost uprooted a segment of compact cameras for casual photography and camera specification has become one of the key considerations for any buyer while selecting a new smartphone. Fig. 1 shows the comparison of number of photos taken with smartphones, digital cameras, and tablets since 2019.

Most important components in a camera

Cameras come in all shapes and sizes. But irrespective of their performance, the digital cameras have three essential parts: A sensor that is the heart of the system, lens or lenses that act as eye(s) of the camera, and the software that converts the analogue data into a usable image. All the three components need to perform well to achieve a good result. But in this article, we are dealing only with the camera sensor.

The camera sensor

A camera sensor is a solid-state device that captures light in small light-sensitive elements called pixels. The sensor detects light as an analogue signal, which is then amplified and converted into a digital signal to store and reproduce later. Sensors cannot produce colours and require a special filter to sense colours.

There are various types of sensors available in the market, such as CCD sensor, CMOS sensor, LiveMOS sensor, or Foveon X3 sensor. The CCD and CMOS sensors are the most common types of sensors in use.

CMOS sensor. Complementary metal oxide semiconductor (CMOS) sensors use rolling shutters and are the most widely used sensors as they can achieve faster data transfer rates (speed) and thus can click pictures at higher frames per second (fps) for a video picture. But as

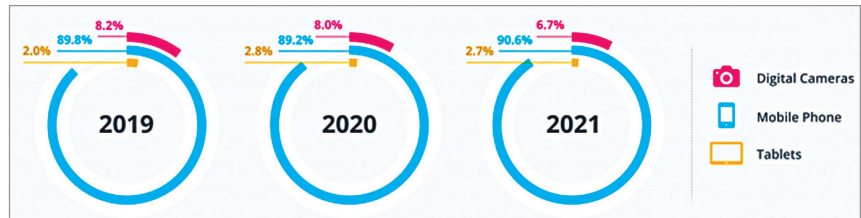


Fig. 1: Statistics of pictures taken with different devices (Source: Link)

the pixels are read in line-by-line fashion, it causes the rolling-shutter effect. However, CMOS sensors do not produce any smear or blooming effect on the pictures. Further, they have higher dynamic range. But these days CMOS sensors, such as Sony's IMX661 sensor, also come with a global shutter.

CCD sensor. Charge coupled device (CCD) sensors use a specialised fabrication technology and do not have a conversion electronic circuit on the sensor surface. The absence of electronics from the surface means that CCD can capture more light and have a lower noise factor. Besides, these sensors have higher colour fidelity, making them a better choice in low-light, low-speed use. These sensors feature a global shutter.

The comparison of CCD sensors with CMOS sensors is given in Table 1.

Applications

Cameras are used in a variety of applications, including monitoring the movement of people or objects in surveillance systems, such as traffic monitoring systems, drones, etc. Cameras are also used for process automation in factories as a part of machine vision systems, or in automobiles for assisting a driver or enabling self-driving of autonomous vehicles. But the most evident use is in our smartphones for capturing moments.

Each of these applications have different requirements. A com-

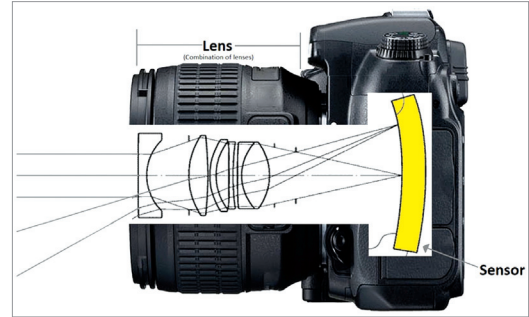


Fig. 2: Path of light falling on the sensor (Source: Link)

TABLE 1
CCD SENSORS VS CMOS SENSORS

Specification	CCD	CMOS
Power consumption	Very high	Low
Cost	High	Low
Speed (fps)	Low	High
Image noise	Low	High
Manufacturing	Difficult	Simple
Uses	Professional cameras	Smartphones, compact cameras, etc
Shutter	Global shutter	Rolling shutter

parison of machine vision camera, surveillance camera, and consumer cameras is given in Table 2.

Imaging sensor parameters

The camera is used in various applications and hence has a lot of different needs and specifications. To understand which imaging sensor is best for our application, we must know some basics about camera sensors. Some important terminologies and technologies are mentioned below.

Pixel and resolution. Before understanding resolution, let us understand what a pixel is. Picture